Providing Location Services within a Radio Cellular Network using Ellipse Propagation Model

Junyang Zhou, Kenneth Man-Kin Chu, Joseph Kee-Yin Ng

Presenter : Junyang Zhou

Department of Computer Science,
Hong Kong Baptist University

Outline

- Introduction
- Related works
- The Ellipse Propagation Model (EPM)
- The Geometric Algorithm based on the EPM
- Simulation Results
- Conclusion and Future work

Introduction

Positioning Technology

Global Positioning System (GPS)

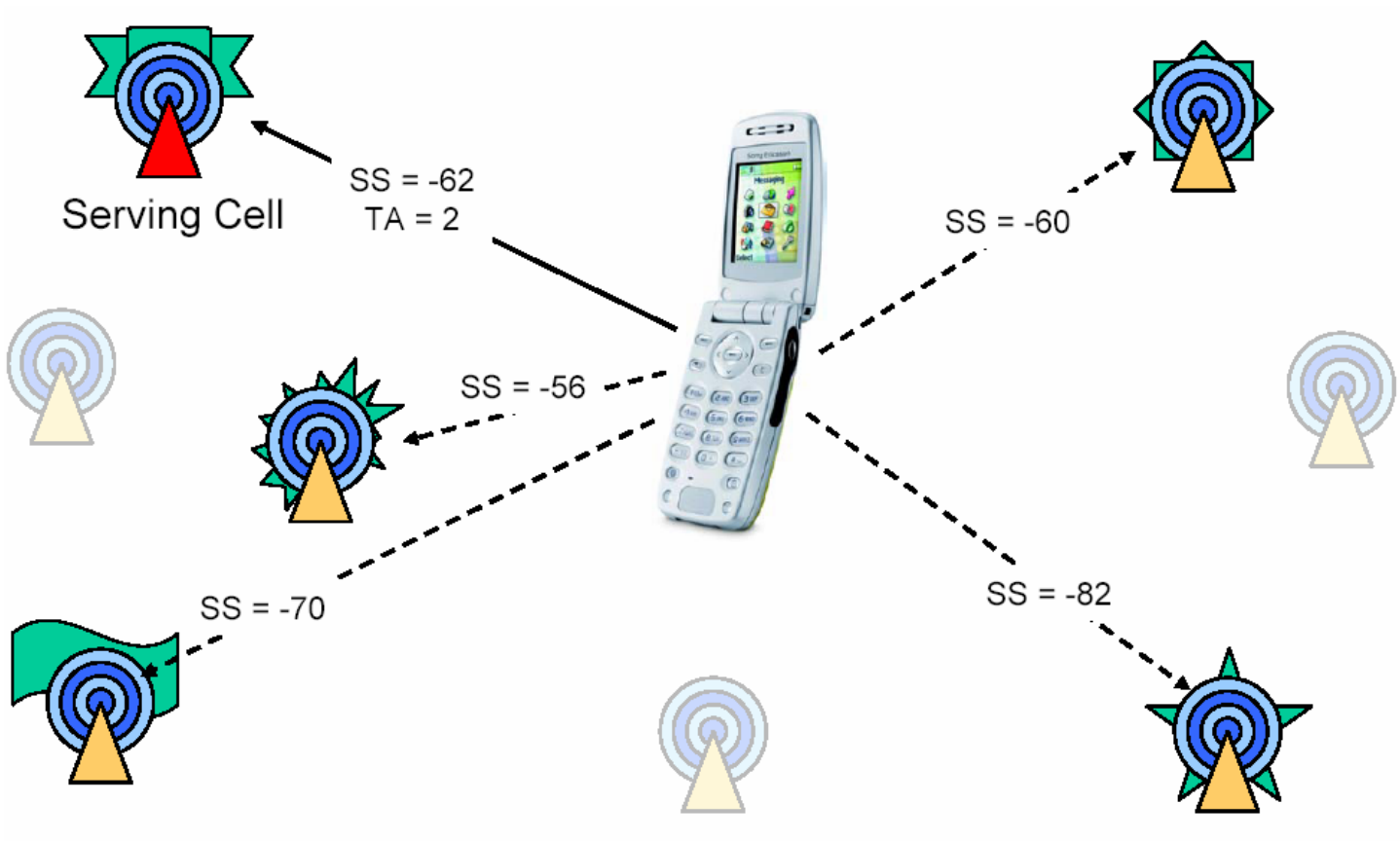
- A GPS receiver measures the distances from 3 or more satellites and utilize a mathematical method, called "Triangulation" to calculate the actual position
- Blind Spot Problem: Signals from the satellites would be blocked, and reflected by tall buildings.
- Compensations
  - Position Reckoning: Electronic Compass, and Speedometer
  - Ground Stations

Positioning Technology

Location Estimation based on Cellular Phone Network

- CDMA: ToA, TDoA, E-OTD, AoA
- GSM: Signal Strength Based Algorithms
- Accuracy depends on the cell type & layout
  - Pico-cell: 10-1000m
  - Micro-cell: 100-1000m
  - Small Macro-cell: 1000-3000m
  - Large Macro-cell: 3000-30Km
- Feasible for Metropolitan areas like Hong Kong

GSM Network



Related works

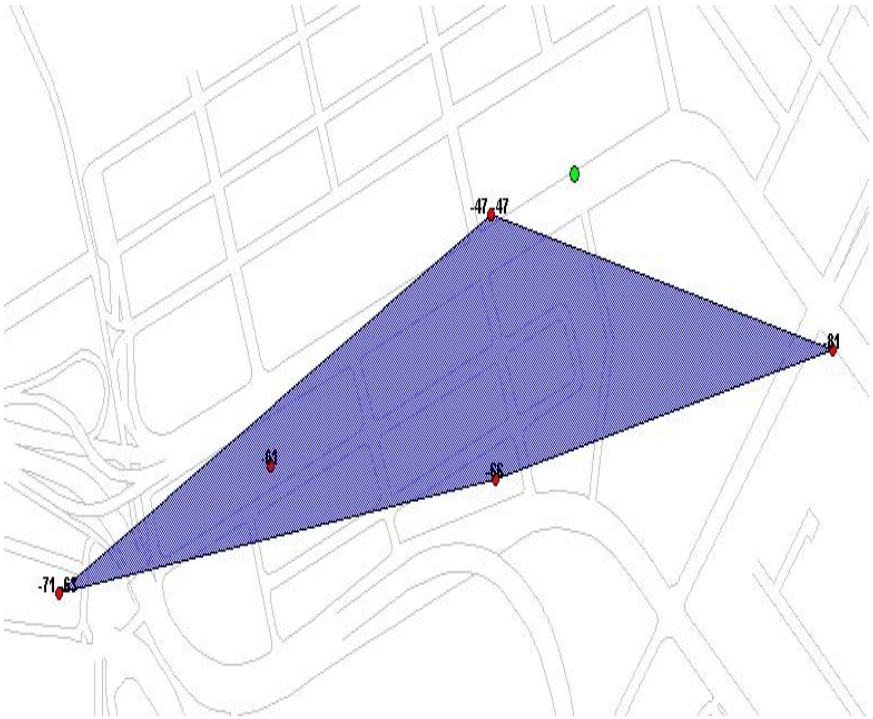
- Center of Gravity(CG)

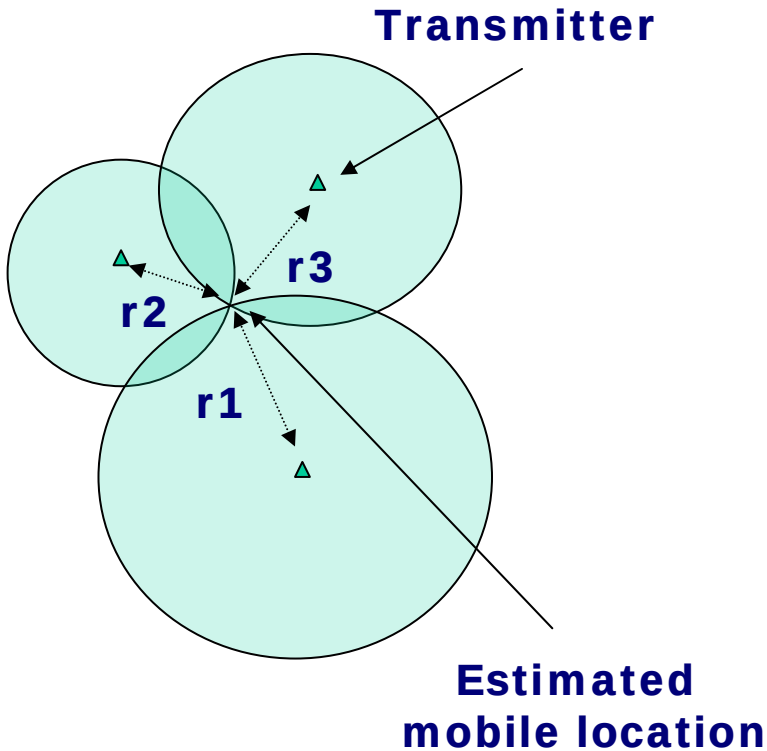
$$\begin{cases} x = \dfrac{x_1 s_1^{-\alpha} + x_2 s_2^{-\alpha} + x_3 s_3^{-\alpha} + \ldots + x_n s_n^{-\alpha}}{s_1^{-\alpha} + s_2^{-\alpha} + s_3^{-\alpha} + \ldots + s_n^{-\alpha}} \\ y = \dfrac{y_1 s_1^{-\alpha} + y_2 s_2^{-\alpha} + y_3 s_3^{-\alpha} + \ldots + y_n s_n^{-\alpha}}{s_1^{-\alpha} + s_2^{-\alpha} + s_3^{-\alpha} + \ldots + s_n^{-\alpha}} \end{cases}$$

- Circular Trilateration(CT)

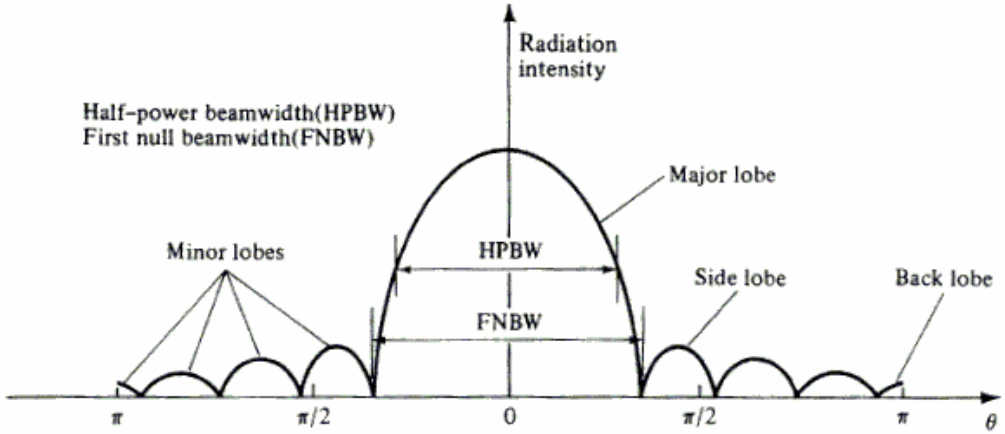
$$\begin{cases} (x - x_1)^2 + (y - y_1)^2 = (\frac{k}{{s_1}^{\alpha}})^2 \\ (x - x_2)^2 + (y - y_2)^2 = (\frac{k}{{s_2}^{\alpha}})^2 \\ (x - x_3)^2 + (y - y_3)^2 = (\frac{k}{{s_3}^{\alpha}})^2 \end{cases}$$

CG & CT

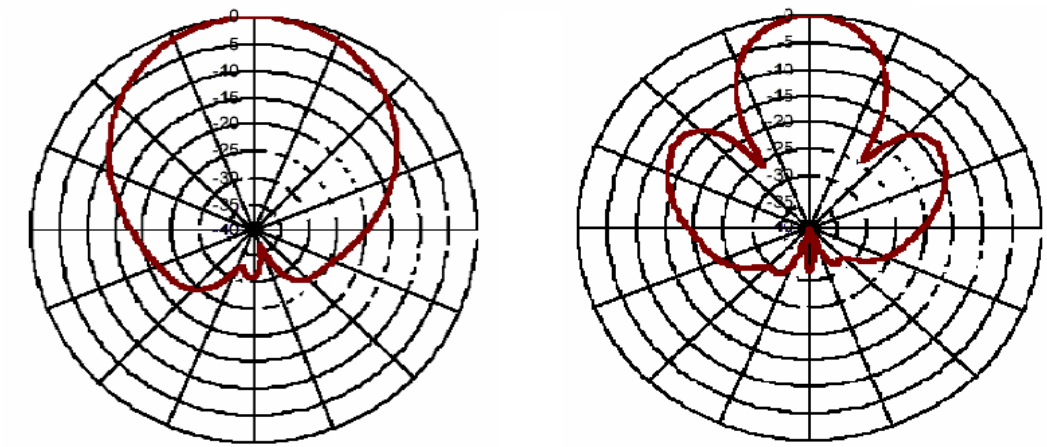

-47
-47
-71
-63
-64
-66
-61


Transmitter
r3
r2
r1
Estimated
mobile location

Directional Transmission Property



Involve Directional Property of All Antenna Types



The Ellipse Propagation Model

- Consider the effect of antenna signal directional transmission
- Suppose the contour line of signal strength is an ellipse
- Consider the base station is on one of the focuses
- Describe the relationship between the signal strength and the distance

The Ellipse Propagation Model

- Translate the signal strength into the distance

The Ellipse Propagation Model:

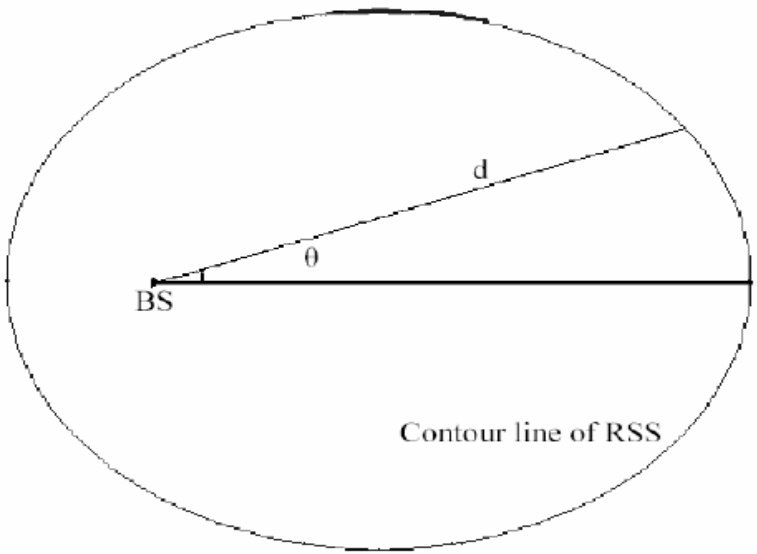
$$d = k(s_0/s)^{1/\alpha}(1-e)/(1-e\cos(\theta))$$

where

θ is the deviation,
s is the signal power received,
s_0 is the transmitting power of the BS,
e is the eccentricity of the ellipse,
k and α are constants,

The Ellipse Propagation Model

- The Geometric Interpretation of the EPM



The Geometric Algorithm

- Use three base stations which are **not in a line** to locate the mobile station
- Use the Ellipse Propagation Model (**EPM**) to translate the signal strength into the distance
- Solve three equation groups derived from these three circles formulas
- Choose the **center** of these three solutions as the estimation of the mobile station

The Geometric Algorithm

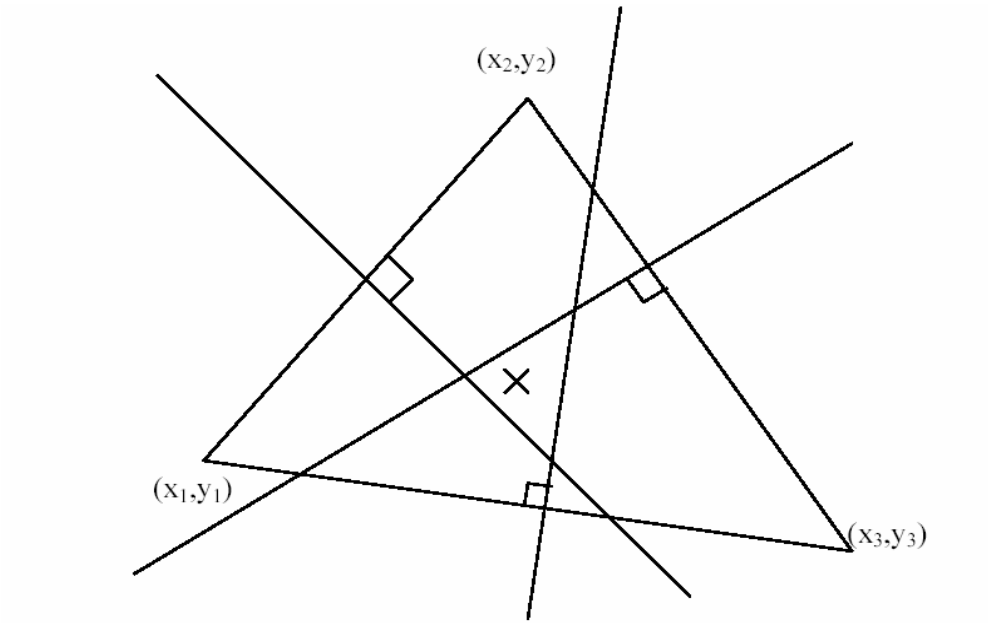
- Three circles formulas

$$d_1^2 = (x - x_1)^2 + (y - y_1)^2$$
$$d_2^2 = (x - x_2)^2 + (y - y_2)^2$$
$$d_3^2 = (x - x_3)^2 + (y - y_3)^2$$

- These three circles can derive three straight lines
- Generate three equation groups from these three straight lines
- Each equation group can only derive one solution
- Choose the **mean** of these three solutions as the estimation

The Geometric Algorithm

- The Geometric Interpretation of the Geometric Algorithm



- These three points on the triangle are the three base stations locations
- Each intersection of two lines is the solution of the equation group
- The "**cross**" point is the estimation of the Geometric Algorithm

The Geometric Algorithm

- Steps of the Geometric Algorithm

Step 1: Given values of parameters of EPM, the RSS, s, and the transmission angle of the BS, θ.

Step 2: Compute the distances between the MS and BSs by the EPM.

Step 3: Compute the estimation of the MS location (x, y) by the Geometric Algorithm.

Step 4: Output (x, y) as the estimation of MS location.

Simulation Results

Lookup table

Three types of models: Macro, Micro, and others

Divide the antennas into three types, each type builds up an ellipse model

Region	e1	e2	e3	alpha	Region	e1	e2	e3	alpha
Aberdeen	0.3	0	0	6.4	ShamTseng	0.9	0.1	0	5.3
CauseWayBay	0.5	0.9	0	4.2	ShaTin	0.5	0.8	0	5.9
Central	0	0.9	0	6.5	ShekKipMeiPark	0.8	0	0	5.8
CheungShaWan	0	0.6	0	8.8	SheungShui	0	0	0	6.9
FoTan	0.9	0	0	6.3	SheungWan	0	0.9	0	7.1
HappyValley	0.7	0	0	6.8	TaiKooShing	0.9	0	0	5.4
HungHom	0.8	0	0	7	TaiWai	0.3	0	0	6.4
KowloonBay	0	0	0.4	8.7	TaiWoHau	0.2	0	0	5.8
KowloonCity	0.6	0.7	0	6.2	TinShuiWai	0.9	0	0	5.6
KowloonTong	0	0	0.4	7.2	ToLoHighway	0.1	0	0	5.4
KwaiFong	0.9	0.9	0	6.1	TsingYi	0.1	0	0	7
KwunTong	0	0	0	8.2	TsuenWan	0.9	0.9	0	8.9
LaiChiKok	0	0	0	7.8	TszWanShan	0.1	0	0	7.1
LaiKing	0.6	0.9	0.5	7.1	TuenMun	0.4	0	0.4	6.4
MaOnShan	0	0	0	8.9	WanChai	0.1	0.9	0.9	7
Mongkok	0.5	0.9	0.1	5.3	WongTaiSin	0.9	0	0	7.2
PE-MK	0.4	0.9	0.1	8.9	YauTong	0.1	0	0	6.2
PrinceEdward	0	0.9	0	7.9	YauYatChuen	0.3	0	0	7
ShamShuiPo	0.4	0.3	0.7	8.9	YuenLong	0.9	0	0	5.3

Result of the Geometric Algorithm

Give the special points values: Mean, Standard deviation, the 67% point value and the 90% point value

Region	Aver.	Std.	67%	90%	Region	Aver.	Std.	67%	90%
Aberdeen	240.55	146.23	247.33	490.56	ShamTseng	1907.11	845.49	2543.35	3027.96
CauseWayBay	256.83	189.26	324.81	651.46	ShaTin	347.48	170.48	395.34	565.30
Central	108.12	64.61	140.43	198.41	ShekKipMeiPark	320.92	186.32	366.48	584.90
CheungShaWan	119.86	67.91	156.51	213.10	SheungShui	492.11	307.42	803.50	888.64
FoTan	251.15	143.00	299.35	465.62	SheungWan	116.48	63.43	151.79	199.32
HappyValley	331.35	207.91	396.63	622.09	TaiKooShing	225.64	146.76	253.14	424.71
HungHom	931.83	530.23	1369.27	1598.5	TaiWai	227.45	90.65	268.78	322.31
KowloonBay	201.45	107.26	233.50	328.65	TaiWoHau	252.71	136.89	287.68	458.90
KowloonCity	220.21	127.16	272.06	503.10	TinShuiWai	375.17	180.10	487.58	641.71
KowloonTong	252.34	155.49	311.68	498.18	ToLoHighway	918.92	614.16	1375.36	2296.17
KwaiFong	187.20	110.77	209.55	347.50	TsingYi	482.61	242.95	587.93	838.96
KwunTong	113.86	60.43	133.78	217.37	TsuenWan	145.52	60.36	159.84	234.84
LaiChiKok	328.71	197.89	403.47	599.42	TszWanShan	246.83	128.49	292.24	441.80
LaiKing	783.46	313.36	976.17	1287.32	TuenMun	334.14	153.98	425.46	520.87
MaOnShan	356.42	122.84	404.56	486.20	WanChai	166.22	109.99	185.72	314.63
Mongkok	95.06	53.59	119.77	166.72	WongTaiSin	260.43	145.77	310.73	473.51
PE-MK	111.69	75.66	138.43	202.52	YauTong	458.96	323.61	537.43	834.18
PrinceEdward	120.89	73.05	136.09	224.27	YauYatChuen	226.48	111.04	274.05	381.47
ShamShuiPo	104.42	58.75	118.45	196.07	YuenLong	238.59	242.99	213.62	769.44

Compare with different algorithms

- Our method has better accuracy than the CG and CT do
- The Geometric Algorithm has the best successful ratio

Model	Average Error	Standard Deviation	sample number	successful %
CG	335.35	319.61	96207	92.6%
CT	321.62	229.06	96207	81.5%
EPM with Geometric Algorithm	285.27	309.80	96207	98.3%

Conclusion and Future work

- Conclusion
  - Yields a high accuracy in Hong Kong
  - Simple and useful

- Future work
  - Extend the 2-D EPM into a 3-D model
  - Develop and Implement some algorithms based on the 3-D model
  - Find out a filter to reduce the effect of signal fluctuation

Thank you!